

Question Paper

Exam Date & Time: 12-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

AI for Medical Image Analysis [CSE-4011]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the Sequential K-Means algorithm to cluster the following 2D dataset with 6 points: (5)

A)

P1	(1,1)
P2	(2,1)
P3	(4,3)
P4	(5,4)
P5	(8,8)
P6	(9,10)

Find two (2) clusters by using P1 and P4 as initial cluster centers.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Analyse how domain knowledge is applied differently in hierarchical segmentation versus foreground segmentation. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Explain the difference between push and pull modes of image transfer and how each affects network efficiency and clinical workflow. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) Describe the main steps in Mahapatra et al.'s active learning process for image segmentation, including how the initial model is trained, how uncertainty and diversity guide sample selection, and how a conditional GAN is used to enhance the labelled dataset before retraining. (4)

A)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) Analyse the importance of annotated imaging datasets in building effective machine learning models for radiology, and explore the factors that influence their reliability and availability. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Analyse the role of intensity comparisons in the BRIEF descriptor and explain how they contribute to (3)
efficient keypoint matching.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) How can domain knowledge about boundary appearance be applied to improve the segmentation (5)
of a foreground object with unclear boundaries in an ultrasound image using methods such as
A) active contours or level sets?

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Apply the concept of temporal continuity to describe how you would segment a 4D (3D+ Time) MRI (5)
heart sequence consistently across multiple time steps.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Analyse how the use of both spatial and contextual encoding paths in the Dual Encoding U-Net (5)
architecture enables the model to balance local detail preservation with global context
A) understanding for effective retinal vessel segmentation.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Analyse how the initial low-resolution classification step in the coarse-to-fine ML-assisted tumour (5)
detection pipeline contributes to the overall efficiency of the framework. Why is it critical to the
system's performance?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Analyse the architectural differences between TensorFlow and Caffe and discuss how these (5)
differences impact their use in designing deep learning models for medical image segmentation.

- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Explain how organizing radiology data in a consistent and standardized format can help in the (3)
development of machine learning applications.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) Explain why a multivariate normal distribution is commonly used for modelling feature vectors. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name			Subject
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025			CSE-4011:AI for Medical Image Analysis
#	Course Outcome	Description	Marks
1	1	CO 1	2
2	2	CO 2	3
3	3	CO 3	20
4	4	CO 4	20
5	5	CO 5	5
Total			50

* **NA** - CO Not Selected

Event Name			Subject
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination -			CSE-4011:AI for Medical Image

Apr-May 2025

Analysis

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	11(22%)
3	Application	3	15(30%)
4	Analysis	6	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	13	50 (100%)